ENGINEERING DESIGN & INNOVATION

MINI PROJECT

MEDICAL STORE MANAGEMENT SYSTEM

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Core Subjects Referenced

Object Oriented Programming • Database Management Systems • Data Structures • Software Engineering • Web Technology • Principles of Programming Languages

Table of Contents

Definitions and Acronyms

Term	Meaning
MSMS	Medical Store Management System
POS	Point of Sale
FEFO	First-Expiry-First-Out
RBAC	Role-Based Access Control
PO	Purchase Order
SKU	Stock Keeping Unit / medicine stock identifier
SRS	Software Requirements Specification
CRUD	Create, Read, Update, Delete
Invoice	Document representing a completed sale
Batch	A specific production lot of a medicine with its own expiry and quantity
Atomic Transaction	A transaction in which all required database updates succeed together or none are committed

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the complete functional, non-functional, interface, data, security, and operational requirements of the Medical Store Management System (MSMS). The document provides a common reference for analysis, design, development, testing, demonstration, and evaluation of the mini project.

MSMS is intended to automate major activities of a medical store, including medicine and batch inventory management, point-of-sale billing, supplier and purchase management, prescription linkage, returns, expiry monitoring, low-stock alerts, customer billing queues, and management reporting. A major design objective is transactional consistency: a completed sale must update the invoice, stock, and financial ledger together, or none of those changes should be committed.

1.2 Project Overview

The system provides a web-based interface through which authorized staff can search medicines, create customer bills, process sales, manage inventory, receive stock, monitor expiry dates, and generate reports. Medicines are maintained at both master and batch levels so that stock quantity and expiry information can be tracked accurately.

During billing, the system checks medicine availability and applicable prescription requirements. For eligible stock, the system follows FEFO (First-Expiry-First-Out), selecting the batch with the earliest valid expiry date before later-expiring batches. The sale is processed atomically so that partial updates are avoided.

1.3 Problem Statement

Manual or loosely integrated medical-store processes can cause billing delays, stock inaccuracies, missed expiry dates, incorrect batch selection, weak purchase tracking, and difficulty tracing prescription-linked sales and returns. The proposed system addresses these problems through a centralized, role-based web application with structured database transactions and automated alerts.

1.4 Objectives

- Automate medicine billing and invoice generation.
- Maintain accurate medicine, category, batch, supplier, and stock information.
- Track expiry dates at batch level and provide expiry alerts.
- Use FEFO logic for stock deduction during sales.
- Maintain a fair and ordered customer billing queue.
- Link prescriptions with sales wherever required by store policy.
- Manage suppliers, purchase orders, and incoming batches.
- Process returns with stock reversal and reason tracking.
- Provide sales, inventory, expiry, and low-stock reports.
- Enforce role-based access and maintain auditable records.
- Apply atomic database transactions to critical sale processing.

1.5 Scope

In scope:

- Authentication and role-based access control.
- Medicine and category master management.
- Batch-wise inventory and expiry tracking.
- Supplier and purchase-order management.
- Point-of-sale billing and customer queue management.
- Prescription capture/linkage.
- Returns and stock reversal.
- Low-stock and expiry alerts.
- Sales, stock and management dashboards.
- Report generation and PDF export.
- Audit logging for critical operations.

Out of scope for the base academic version:

- Direct integration with government medicine-regulation portals.
- Online payment gateway settlement.
- Automated insurance claim processing.
- Hospital/doctor information-system integration.
- Autonomous medical diagnosis or prescription generation.

2. Overall Description

2.1 Product Perspective

MSMS is a centralized web application consisting of a presentation layer, application/business logic layer, and relational database layer. The presentation layer provides role-specific screens; the business layer implements validation, billing, FEFO, queue handling, purchase, return and reporting logic; and the database layer stores persistent system records.

2.2 Product Functions

Function Area	Main Capabilities
Authentication	Login, logout, sessions/tokens, password protection, role checks.
Medicine Management	Create/update/deactivate medicine records, categories and reorder levels.
Supplier Management	Supplier records, contact details, purchase history.
Purchasing	Create POs, receive stock, create batches, record expiry and purchase cost.
Billing	Queue customer, search medicines, cart, prescription linkage, invoice.

Inventory	Batch quantity, FEFO deduction, low-stock and expiry monitoring.
Returns	Return entry, validation, stock reversal, reason and audit record.
Reporting	Sales, stock, expiry, low-stock and purchase reports; PDF export.
Administration	Users, roles, configuration, audit records and dashboard.

2.3 Operating Environment

- Client: modern web browser on desktop/laptop or suitable tablet.
- Server: application server capable of hosting the selected web technology.
- Database: relational database such as MySQL.
- Network: local network or secure internet connection depending on deployment.
- Development environment: IDE/editor, version control, database administration tool and browser developer tools.

2.4 Design Principles

- Modularity: each major business capability is separated into manageable modules.
- Encapsulation: business rules are implemented behind well-defined service interfaces.
- Consistency: critical inventory and billing updates use database transactions.
- Usability: billing should require minimal unnecessary navigation.
- Security: users receive only permissions required for their role.
- Traceability: critical actions should be associated with a user and timestamp.
- Maintainability: code, database schema and interfaces should be organized for future extension.

3. Stakeholders and User Classes

Stakeholder / Role	Responsibilities	Access Level
Cashier / Pharmacist	Customer queue, medicine search, billing, prescription linkage, returns as permitted.	Operational
Store Manager	Inventory, purchases, suppliers, reports, expiry/low-stock monitoring.	Management
Administrator	Users, roles, system configuration, audit monitoring and full management access.	Full
Customer	Receives invoice and service; does not directly administer the system.	Indirect
System Developer / Maintainer	Deployment, maintenance, backup and technical troubleshooting.	Technical

3.1 User Characteristics

Operational users are expected to have basic computer literacy and familiarity with medicine-store billing. Managers and administrators require additional knowledge of stock, purchase and reporting operations. The system should minimize technical complexity and provide clear validation messages.

4. Functional Requirements

ID	Requirement	Description
FR-01	User Authentication	The system shall authenticate users using valid credentials before allowing access to protected functions.
FR-02	Role Authorization	The system shall restrict features according to Cashier/Pharmacist, Store Manager and Admin roles.
FR-03	Medicine CRUD	Authorized users shall add, view, update and deactivate medicine master records.
FR-04	Category Management	Authorized users shall create and maintain medicine categories and subcategories.
FR-05	Reorder Level	The system shall store a configurable reorder level for each medicine.
FR-06	Supplier Management	Authorized users shall maintain supplier details and supplier status.
FR-07	Purchase Order	Managers shall create, edit, approve as applicable, and track purchase orders.
FR-08	Batch Receiving	The system shall record batch number, expiry date, received quantity, purchase price and supplier for incoming stock.
FR-09	Inventory View	Authorized users shall view medicine stock by medicine and batch.
FR-10	Expiry Tracking	The system shall identify expired and soon-to-expire batches using configurable alert thresholds.
FR-11	Low Stock Alert	The system shall identify medicines whose usable stock

		is at or below their reorder level.
FR-12	Customer Queue	The system shall assign ordered queue numbers/tokens to customers waiting for billing.
FR-13	Medicine Search	The billing screen shall support search by medicine name and configured identifiers.
FR-14	Cart Management	The cashier shall add, modify and remove sale items before checkout.
FR-15	Prescription Linkage	The system shall allow a prescription record to be linked to a sale when required by configured store rules.
FR-16	Stock Validation	The system shall prevent checkout when sufficient valid stock is unavailable.
FR-17	FEFO Deduction	The system shall deduct sale quantities from eligible batches using earliest-expiry-first ordering.
FR-18	Atomic Sale	The system shall commit invoice, invoice items, stock deduction and ledger entry as one database transaction.
FR-19	Invoice Generation	The system shall generate a unique invoice for every completed sale.
FR-20	Returns	Authorized users shall record eligible returns, reasons and quantities and reverse stock according to policy.
FR-21	Reports	The system shall generate sales, stock, expiry, low-stock and purchase reports.
FR-22	Dashboard	The system shall provide role-appropriate summary information and alerts.
FR-23	Audit Log	The system shall record selected critical actions with user, timestamp and action information.
FR-24	PDF Export	Authorized users shall export supported reports/invoices to PDF.

FR-25	Logout	The system shall terminate the authenticated session securely when the user logs out.

4.1 Authentication and Access Control

- The login page shall accept the configured username/email and password.
- Invalid credentials shall not grant access and shall produce a generic error message.
- A successful login shall create an authenticated session/token.
- Every protected request shall be authorized against the user's role.
- Logout shall invalidate the active session/token.
- Administrative functions shall not be available to operational roles.

4.2 Medicine and Category Management

Each medicine master record shall contain at minimum a unique identifier, medicine name, category, unit/pack information, selling price, reorder level and active/inactive status. Batch information shall be stored separately so multiple batches can exist for the same medicine.

4.3 Supplier and Purchase Management

The purchase module shall support supplier selection, purchase order creation, item quantities, expected delivery information, receiving status, and creation of batch-level stock when goods are received. Received stock shall increase inventory only after a valid receiving operation.

4.4 Billing / Point of Sale

A cashier shall select the next customer in the queue, search medicines, add quantities to a cart, review prices and applicable adjustments, and submit checkout. The system shall validate each line item before committing the sale.

Checkout sequence:

1. Identify the active customer/queue token.
2. Validate the cart and required fields.
3. Validate prescription linkage where applicable.
4. Find eligible non-expired stock batches.
5. Allocate quantities using FEFO.
6. Calculate totals and prepare invoice data.
7. Start a database transaction.
8. Lock/revalidate relevant stock records to prevent conflicting deductions.
9. Create invoice and invoice-item records.
10. Deduct allocated batch quantities.
11. Create corresponding ledger/transaction entry.
12. Commit the transaction.
13. Remove/complete the customer from the active billing queue.
14. Generate/display the invoice.

4.5 Stock and Expiry Tracking

Stock shall be represented at batch level. A batch shall include a batch identifier, medicine reference, expiry date, received quantity, available quantity, purchase price and receiving information. Expired stock shall not be automatically offered for normal sale. The system shall support configurable warning periods for upcoming expiry.

4.6 Prescription Management

The prescription module shall store a reference record and, where permitted by the project deployment, an uploaded prescription document. The sale shall retain a link to the prescription record. The system shall not attempt to diagnose a customer or generate medical advice.

4.7 Returns Processing

A return shall identify the original invoice where possible, the returned medicine/quantity, reason, date, operator and disposition. If the returned item is eligible for restocking under store policy, inventory shall be increased through a controlled reversal. Otherwise, the return shall be recorded without making the quantity available for sale.

5. Non-Functional Requirements

ID	Quality Attribute	Requirement
NFR-01	Performance	Normal medicine searches and common screen operations should respond promptly under expected mini-project load.
NFR-02	Reliability	Critical billing operations shall avoid partial database updates.
NFR-03	Availability	The deployed system should be available during store operating hours, subject to infrastructure.
NFR-04	Security	Passwords shall be protected using secure password hashing; protected functions require authentication and authorization.
NFR-05	Usability	Billing workflow shall be simple, readable and require minimal unnecessary clicks.
NFR-06	Maintainability	Modules shall be separated with clear responsibilities and reusable components.
NFR-07	Scalability	The architecture should allow additional stores, users and reports to be added with limited redesign.

NFR-08	Data Integrity	Database constraints and transactions shall prevent invalid references and inconsistent stock.
NFR-09	Auditability	Critical operations shall have traceable user/time information.
NFR-10	Portability	The web interface should work on commonly used modern browsers.
NFR-11	Backup	Database backup procedures shall be defined for deployed environments.
NFR-12	Privacy	Prescription and customer-related records shall be accessible only to authorized users.

6. External Interface Requirements

6.1 User Interface

Screen	Required Elements
Login	Username/email, password, login action, validation messages.
Dashboard	KPIs, low-stock alerts, expiry alerts, quick actions and role-specific summaries.
Medicine	Search, filters, medicine list, add/edit/deactivate controls.
Suppliers	Supplier list, details, status, purchase history.
Purchase	PO creation, supplier selection, items, quantities, receiving and batch details.
Billing	Queue, medicine search, cart, prescription indicator, totals, checkout.
Inventory	Medicine/batch table, quantity, expiry, status and alerts.
Prescription	Reference details, upload/link field, associated sale information.
Returns	Invoice lookup, item selection, quantity, reason and confirmation.
Reports	Report type, date/filter controls, table/summary and PDF export.
Users/Admin	User list, roles, status, audit information and system settings.

6.2 Software Interfaces

- Web browser communicates with the application server over HTTP/HTTPS.
- Application communicates with a relational database through a database access layer.
- PDF generation component may be used for invoices and reports.
- File storage may be used for prescription documents subject to deployment policy.

6.3 Hardware Interfaces

The base system requires a computer or tablet capable of running a modern web browser. Optional hardware such as a barcode scanner or receipt printer may be integrated in future versions.

7. System Architecture and Design

7.1 Proposed Three-Layer Architecture

Layer	Responsibilities
Presentation Layer	Web pages/components, forms, dashboards, client-side validation and user interaction.
Business/Application Layer	Authentication, authorization, billing, FEFO allocation, purchase, returns, reports and business rules.
Data Layer	Repositories/DAO, SQL queries, transactions, constraints and persistent storage.

7.2 High-Level Processing Flow

User Login → Role Verification → Dashboard → Select Module → Validate Input → Execute Business Logic → Database Transaction/Query → Update Records → Return Result → Display Confirmation/Report.

7.3 OOP Design Considerations

The implementation should model important entities such as User, Role, Medicine, Category, Batch, Supplier, PurchaseOrder, PurchaseItem, CustomerQueue, Prescription, Invoice, InvoiceItem, Return and AuditLog. Services should encapsulate business rules, while repository/data-access classes should isolate persistence.

8. Data Requirements and Database Design

8.1 Main Entities

Entity	Purpose
User	Stores authenticated user information and role reference.
Role	Defines access category and permissions.
Medicine	Stores medicine master information.

Category	Groups medicines.
Batch	Stores batch number, expiry and quantity for a medicine.
Supplier	Stores supplier information.
PurchaseOrder	Header record for supplier purchase.
PurchaseItem	Medicine/batch receiving details for a purchase.
CustomerQueue	Stores billing queue token/status.
Prescription	Stores prescription reference and linkage information.
Invoice	Sale header containing customer, operator, totals and timestamp.
InvoiceItem	Individual medicine lines within an invoice.
SaleAllocation	Records which batch supplied each sold quantity.
Return	Return header and reason information.
ReturnItem	Returned medicine and quantity.
LedgerEntry	Financial transaction reference associated with a sale/return.
AuditLog	Records selected user actions for traceability.

8.2 Suggested Relational Schema

Table	**Important Fields**
roles	role_id PK, role_name, description
users	user_id PK, role_id FK, name, username, password_hash, status, created_at
categories	category_id PK, name, description, status
medicines	medicine_id PK, category_id FK, name, generic_name, unit, selling_price, reorder_level, status
batches	batch_id PK, medicine_id FK, batch_no, expiry_date, received_qty, available_qty, purchase_price, received_at
suppliers	supplier_id PK, name, contact, phone, email, address, status
purchase_orders	po_id PK, supplier_id FK, order_date, status, total_amount
purchase_items	purchase_item_id PK, po_id FK, medicine_id FK, batch_id FK, quantity, purchase_price
queue_entries	queue_id PK, token_no, customer_label, status, created_at, served_at
prescriptions	prescription_id PK, reference_no, file_path/reference, captured_by, captured_at
invoices	invoice_id PK, invoice_no, queue_id FK, user_id FK, prescription_id FK nullable, total, status, created_at

invoice_items	invoice_item_id PK, invoice_id FK, medicine_id FK, quantity, unit_price, line_total
sale_allocations	allocation_id PK, invoice_item_id FK, batch_id FK, quantity
returns	return_id PK, invoice_id FK, user_id FK, reason, total, created_at
return_items	return_item_id PK, return_id FK, medicine_id FK, batch_id FK nullable, quantity, restock_status
ledger_entries	ledger_id PK, reference_type, reference_id, amount, entry_type, created_at
audit_logs	audit_id PK, user_id FK, action, entity_type, entity_id, timestamp, details

8.3 Database Integrity Rules

- Primary keys shall uniquely identify records.
- Foreign keys shall preserve valid relationships.
- Medicine and category records required for active operations shall not be deleted in a way that breaks historical records; deactivation is preferred.
- Batch available quantity shall not become negative.
- Invoice numbers shall be unique.
- Batch expiry date shall be validated against receiving/sale rules.
- Critical sale updates shall be executed inside a database transaction.
- Timestamps shall be recorded for important operational records.

9. Business Rules and Processing Logic

9.1 FEFO Rule

For a requested medicine quantity Q, the system shall consider only active batches with available quantity greater than zero and an expiry date that makes the batch eligible for sale. Eligible batches shall be ordered by expiry date ascending. The system shall allocate quantity from the earliest-expiring batch first, continuing to the next batch if required.

Example: If Batch A expires in March and Batch B expires in May, a sale shall consume Batch A before Batch B.

9.2 Atomic Sale Transaction

The sale operation shall be treated as an all-or-nothing transaction. If invoice creation succeeds but a required stock deduction or ledger entry fails, the entire transaction shall be rolled back. The system shall not leave a completed invoice without the corresponding stock deduction.

9.3 Concurrency Control

When multiple billing terminals can access the same stock, the implementation should recheck or lock the relevant batch rows during allocation/update so two simultaneous sales cannot consume the same available quantity.

9.4 Low-Stock Rule

A medicine is considered low-stock when its usable aggregate available quantity is less than or equal to its configured reorder level. The dashboard shall display an alert for authorized users.

9.5 Expiry Rule

A batch whose expiry date has passed shall be marked expired and excluded from normal sale allocation. A configurable warning window may identify batches approaching expiry.

9.6 Queue Rule

Customers shall be served in queue order by default. Each active queue entry receives a unique token. The system shall allow authorized staff to mark a token as called, serving, completed or cancelled.

10. Use Cases

ID	Use Case	Actor	Goal
UC-01	Login	All roles	Authenticate and enter the system.
UC-02	Manage Medicines	Manager/Admin	Create, update, deactivate and search medicines.
UC-03	Manage Suppliers	Manager/Admin	Maintain supplier information.
UC-04	Create Purchase Order	Manager	Create and track supplier orders.
UC-05	Receive Stock	Manager	Receive a purchase and record new batches.
UC-06	Monitor Inventory	Manager/Admin	View batch stock, low-stock and expiry status.
UC-07	Add Customer to Queue	Cashier/Pharmacist	Create queue token.
UC-08	Process Sale	Cashier/Pharmacist	Build cart, validate, allocate FEFO stock and complete invoice.
UC-09	Link Prescription	Cashier/Pharmacist	Attach prescription reference/document to a sale where required.
UC-10	Process Return	Authorized user	Record return and controlled stock reversal.
UC-11	Generate Reports	Manager/Admin	View and export reports.

UC-12	Manage Users	Admin	Create/disable users and assign roles.
UC-13	Review Audit Log	Admin	Review critical system actions.

10.1 Detailed Use Case – Process Sale

Field	Specification
Primary Actor	Cashier / Pharmacist
Precondition	User is authenticated; queue customer is active; required medicines are configured.
Trigger	Cashier starts checkout.
Main Flow	Select queue token → search medicine → add quantity → validate stock → validate prescription if required → calculate total → confirm → atomic transaction → invoice.
Alternative Flow	Insufficient stock → show shortage and prevent checkout; expired batch → exclude from allocation; missing prescription → request required linkage.
Postcondition	Invoice is stored, allocated stock is deducted, ledger is recorded, and queue entry is completed.
Failure Condition	If any critical transaction step fails, database changes are rolled back.

10.2 Detailed Use Case – Receive Stock

Field	Specification
Primary Actor	Store Manager
Precondition	Supplier and purchase order exist or direct receiving is authorized.
Main Flow	Select PO → verify items → enter batch number and expiry → enter received quantity → validate → save batch → update purchase status.
Postcondition	New/updated batch stock is available for inventory monitoring.
Validation	Quantity must be positive; batch and expiry information must be valid; medicine must exist.

11. Data Flow and Process Specifications

11.1 Context-Level Data Flow

External entities: Cashier/Pharmacist, Store Manager, Administrator, Supplier and Customer. The MSMS receives login data, medicine/purchase/sale/return information and produces invoices, alerts, dashboards and reports. Supplier information enters through purchase management, while customer interaction is represented through the billing queue and invoice process.

11.2 Level-1 Logical Processes

Process	Input	Output	Primary Data Stores
P1 Authentication	Credentials	Session/authorization result	Users, Roles
P2 Master Management	Medicine/category/supplier data	Updated master records	Medicines, Categories, Suppliers
P3 Purchasing	PO and receiving details	PO status, batch stock	Purchase Orders, Purchase Items, Batches
P4 Billing	Queue/cart/prescription	Invoice, stock update	Queue, Invoices, Invoice Items, Sale Allocations, Batches
P5 Returns	Invoice/return data	Return record, stock reversal	Returns, Return Items, Batches
P6 Reporting	Filters/date ranges	Reports/dashboard	Invoices, Batches, Purchases, Returns, Ledger
P7 Administration	User/config/audit actions	Updated access and logs	Users, Roles, Audit Logs

12. Security and Privacy Requirements

- Passwords shall never be stored in plaintext; a strong password hashing mechanism shall be used.
- Authentication state shall be protected and expired according to deployment configuration.
- Authorization shall be enforced server-side; hiding a UI button alone shall not constitute access control.
- Prescription records shall be accessible only to authorized personnel.
- Input validation shall be applied at both client and server layers.
- Database queries shall use parameterized statements or an ORM/query mechanism that prevents SQL injection.
- Critical actions such as sale, return, stock receiving and user-role changes should be auditable.
- Sensitive documents should use controlled storage and access rather than public URLs.
- HTTPS should be used for deployed environments.
- Regular database backups should be maintained according to deployment requirements.

13. Error Handling and Validation

Situation	System Response

Invalid login	Reject access and show a generic authentication error.
Missing required field	Highlight field and request correction.
Duplicate medicine/batch identifier	Reject or request correction according to uniqueness rule.
Insufficient stock	Prevent checkout and show available quantity.
Expired batch	Exclude from sale and show expiry warning where appropriate.
Missing required prescription	Prevent applicable sale until required linkage is supplied.
Database failure during sale	Rollback critical transaction and show failure message.
Invalid return quantity	Reject and display allowed return quantity/policy.
Unauthorized operation	Reject request and record security event where configured.
File upload failure	Reject incomplete/unsupported upload and preserve sale integrity.

14. Reports and Dashboard Requirements

Report	Contents / Filters
Daily/Periodic Sales	Invoice count, sales total, date range, operator and summary.
Medicine Stock	Medicine, batch, available quantity, expiry, reorder status.
Expiry Report	Expired and soon-to-expire batches, configurable date window.
Low Stock Report	Medicines at/below reorder level.
Purchase Report	Supplier, PO, date, items, quantities and purchase values.
Returns Report	Return date, invoice, medicine, quantity, reason and amount.
Ledger Summary	Sales/return transaction references and amounts.
Audit Report	Critical actions by user/date/entity.

14.1 Dashboard KPIs

- Today's sales amount and invoice count.
- Current low-stock medicine count.
- Expired batch count.
- Upcoming expiry batch count.
- Pending purchase orders.
- Queue length/current token.
- Recent transactions.

15. Technology and Implementation Requirements

The exact implementation stack may be selected by the project team, but the system should follow a web-based architecture and use a relational database. The following is a suitable academic stack.

Layer	Recommended Technology	Purpose
Frontend	HTML5, CSS3, JavaScript; optionally Bootstrap	Responsive web interface and user interaction.
Backend	PHP or Java-based web backend	Business logic, authentication, APIs and transactions.
Database	MySQL	Relational persistent storage.
Database Tool	phpMyAdmin / MySQL Workbench	Schema and database administration.
Web Server	Apache/XAMPP or equivalent	Local/development hosting when using PHP.
PDF	Server-side PDF library	Invoice/report PDF generation.
Version Control	Git/GitHub	Source-code versioning and collaboration.

If Java is selected for the backend, OOP concepts such as classes, interfaces, encapsulation, inheritance where justified, exception handling and service/repository separation can be demonstrated explicitly.

16. Testing Requirements and Acceptance Criteria

16.1 Testing Levels

- Unit Testing: validate individual services, validators and utility functions.
- Integration Testing: validate interaction between business logic and database.
- System Testing: validate complete workflows such as purchase-to-stock and cart-to-invoice.
- Security Testing: verify role restrictions and invalid access attempts.
- Usability Testing: verify billing workflow and readable validation messages.
- Performance Testing: evaluate common operations under expected mini-project workload.

16.2 Sample Test Cases

ID	Test	Input/Condition	Expected Result	Status
TC-01	Valid login	Correct credentials	User reaches role dashboard	Pass/Fail
TC-02	Invalid login	Incorrect password	Access denied	Pass/Fail
TC-03	Medicine search	Existing medicine name	Matching medicines displayed	Pass/Fail
TC-04	Low-stock alert	Stock <= reorder level	Alert appears	Pass/Fail

TC-05	Expiry exclusion	Expired batch exists	Expired batch is not allocated to normal sale	Pass/Fail
TC-06	FEFO	Two eligible batches with different expiry dates	Earlier expiry batch is allocated first	Pass/Fail
TC-07	Insufficient stock	Cart quantity > usable stock	Checkout blocked	Pass/Fail
TC-08	Atomic sale	Force failure during critical update	All critical changes roll back	Pass/Fail
TC-09	Prescription rule	Medicine requiring prescription without linkage	Checkout blocked/configured warning	Pass/Fail
TC-10	Return	Valid invoice return	Return saved and eligible stock reversed	Pass/Fail
TC-11	Role access	Cashier attempts admin action	Access denied	Pass/Fail
TC-12	Report export	Select report and export PDF	Readable PDF generated	Pass/Fail

16.3 Acceptance Criteria

- All critical functional requirements are implemented and demonstrated.
- A complete sale creates a consistent invoice, stock deduction and ledger record.
- A failed sale does not leave partial critical updates.
- FEFO batch allocation is demonstrated with at least two batches.
- Expired stock is excluded from normal sales.
- Low-stock and expiry alerts are visible to authorized users.
- Role-based restrictions work correctly.
- Reports produce correct data for selected date/filter ranges.
- The database maintains valid relationships and prevents negative available stock.
- Core workflows pass documented system tests.

17. Traceability to Core Subjects

Core Subject	Application in Project
Object Oriented Programming	Classes/entities, encapsulation, interfaces, services, exception handling and modular design.
Database Management Systems	Relational schema, primary/foreign keys, normalization, transactions, constraints, SQL and indexing.

Data Structures	Queue for customer billing order; lists/maps for cart and lookup operations; sorting/priority logic for expiry-based allocation.
Software Engineering	SRS, modular architecture, requirements, use cases, testing, validation, maintenance and traceability.
Web Technology	Browser-based UI, forms, client-server communication, authentication and responsive dashboards.
Principles of Programming Languages	Syntax/semantics awareness, data types, control structures, modularity, abstraction and exception/error handling in implementation.

18. Constraints, Assumptions and Dependencies

18.1 Constraints

- The project is an academic mini project and may use a simplified regulatory model.
- The system depends on correct data entry by authorized staff.
- Actual legal/regulatory requirements may vary by jurisdiction and medicine category.
- The base version assumes a stable database and network environment.

18.2 Assumptions

- Every medicine has a unique internal identifier.
- Each physical stock lot can be represented as a batch.
- Authorized staff are responsible for verifying prescriptions where required.
- Managers maintain accurate reorder levels and supplier information.
- The system clock is correctly configured for timestamped records.

18.3 Dependencies

- Relational database availability.
- Web/application server availability.
- Configured user accounts and roles.
- PDF generation component for export.
- Secure storage if prescription documents are uploaded.

19. Future Enhancements

- Barcode/QR-based medicine scanning.
- Multi-branch medical store support.
- Mobile-friendly staff application/PWA.
- Online customer ordering with pharmacist verification.

- Supplier portal and automated purchase suggestions.
- Advanced analytics and demand forecasting.
- Notification integration for stock/expiry alerts.
- Receipt-printer and barcode-printer integration.
- Stronger document retention and compliance workflows.
- Integration with approved external pharmacy/healthcare systems where legally and technically appropriate.

20. Conclusion

The Medical Store Management System provides a structured software solution for automating core medical store operations. Its design combines point-of-sale billing, batch-wise inventory, FEFO stock allocation, supplier and purchase management, prescription linkage, returns, reporting, queue handling and role-based security in one integrated system. The emphasis on database transactions and data integrity makes the project particularly suitable for demonstrating practical DBMS and software engineering concepts.

The SRS establishes a clear foundation for subsequent design, implementation, testing and demonstration. The modular architecture allows the project team to implement a functional academic prototype while retaining a path toward future features such as barcode scanning, multi-store support, analytics and external system integration.

Appendix A – Glossary

Term	Definition
Batch-wise Inventory	Stock tracked separately for each production batch.
FEFO	Method of issuing stock with the earliest expiry first.
Reorder Level	Configured stock threshold at which replenishment is indicated.
Atomicity	A transaction property ensuring all critical operations succeed together or none do.
Role	A named access category assigned to a user.
Audit Log	Historical record of selected actions performed in the system.
Prescription Linkage	Association between a prescription record and a sale.

Appendix B – Sample Data Dictionary

Field	Type (Example)	Description	Validation

medicine_id	INT	Unique medicine identifier	Primary key, positive
medicine_name	VARCHAR	Medicine display name	Required
category_id	INT	Medicine category reference	Foreign key
batch_no	VARCHAR	Production batch number	Required within medicine context
expiry_date	DATE	Batch expiry date	Valid date
available_qty	DECIMAL/INT	Usable stock quantity	>= 0
reorder_level	DECIMAL/INT	Low-stock threshold	>= 0
selling_price	DECIMAL	Sale price per configured unit	>= 0
invoice_no	VARCHAR	Unique sale invoice number	Unique, required
user_id	INT	Operator/admin reference	Foreign key
prescription_id	INT	Prescription reference	Nullable unless required
created_at	DATETIME	Record creation timestamp	System generated

Appendix C – Suggested Project Deliverables

- SRS document.
- Use-case diagram.
- Context diagram and Level-1 DFD.
- ER diagram.
- Database schema/SQL script.
- UI wireframes or prototype.
- Working web application.
- Test-case document and test results.
- Project presentation/demo.
- Source code and version-control repository.